

Sequence representations

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Augmentation and leakage

Without opening the old lesson, explain one key idea from Augmentation and leakage. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Pooling and spatial information

Without opening the old lesson, explain one key idea from Pooling and spatial information. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Learned representations

Without opening the old lesson, explain one key idea from Learned representations. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Expose an order collision for Sequence representations. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Sequence representations. Proposed explanation: “Larger token IDs mean more of a concept”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a fresh schema independently

Use PAD = 0, UNK = 1, OPEN = 2, CLOSE = 3, CHECK = 4 and five stored positions. Encode OPEN CHECK CLOSE and give its mask and final real position. Separately, sensor values [0, 6, 3, 0, 0] have that same mask. Calculate their masked mean and the incorrect mean over all five slots.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Encode and recover the final event

Encode GO TURN STOP using the lesson vocabulary and four positions. Give its mask, real length and final real event.

My reasoning and evidence

6. W2 Expose an order collision

Give two logs made of exactly GO, TURN and STOP once each that have equal count vectors but different answers to whether the final event is STOP.

My reasoning and evidence

7. W3 Separate unknown from padding

Encode WAIT GO. Which positions are real? Compare it with PAUSE GO, and state what information the encoding loses.

My reasoning and evidence

8. W4 Apply a fresh schema independently

Use PAD = 0, UNK = 1, OPEN = 2, CLOSE = 3, CHECK = 4 and five stored positions. Encode OPEN CHECK CLOSE and give its mask and final real position. Separately, sensor values [0, 6, 3, 0, 0] have that same mask. Calculate their masked mean and the incorrect mean over all five slots.

My reasoning and evidence

9. W5 Audit length and empty input

The encoder receives five events ending in STOP, then an empty log. What should it do under the lesson contract? Why should a program not simply crop the first and assume a final event for the second?

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled sequence representations evidence sheet to Image-pattern Network Investigation. Show how you can encode an ordered sequence with an explicit vocabulary. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Sequence representations

Distinguish padding positions from meaningful zero sensor readings.

Toy tokens PAD0, UNK1, LEFT2, RIGHT3; stored length4. Sequence LEFT RIGHT encodes [2, 3, 0, 0], mask [1, 1, 0, 0].

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Sequence representations

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Encode real tokens

LEFT and RIGHT occupy positions 0 and 1.



2. Mark padding

Positions 2 and 3 are storage padding; mask marks them 0.



3. Compare sensor example

Readings [0, 4, 0, 0] with real length 2 use mask [1, 1, 0, 0]; the first zero is a real measurement.

CHANGE ONE THING

A missing sequence length causes every numeric 0 to be treated as padding in the sensor example.

Explain the changed result using the matching lesson.

Which record lets the program distinguish the first and last zeros? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

13. Project portfolio milestone

Project: Image-pattern Network Investigation

Current milestone: Prepare the demonstration. Gather reproducible instructions, evidence, access checks and remaining limitations.

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Distinguish padding positions from meaningful zero sensor readings.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Readings [0, 4, 0, 0] with real length 2 use mask [1, 1, 0, 0]; the first zero is a real measurement.

1. Encode real tokens
2. Compare sensor example
3. Mark padding

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Compare sensor example Readings [0, 4, 0, 0] with real length 2 use mask [1, 1, 0, 0]; the first zero is a real measurement. Sequence values and sequence length are separate information. Padding fills unused storage positions so arrays can share a size; a mask marks which positions are real. A real sensor reading can be zero, just like a padding value. Keep the length or mask rather than inferring presence from the number alone. Token IDs are identifiers, not measured magnitudes.

Practice 2: explain the deciding evidence

Changed-case practice: A missing sequence length causes every numeric0 to be treated as padding in the sensor example. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. The first real sensor reading is incorrectly discarded.

3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The first real sensor reading is incorrectly discarded. Its numeric value is 0, but its position is real according to the declared length.

Repair a missing building block

Review Augmentation and leakage (ai-5/24). Goal: Check whether a transformation preserves the task label. Short practice: How many independent source groups are in the final test? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Pooling and spatial information (ai-5/22). Goal: Compute max and mean pooling under an explicit window rule. Short practice: Can the max map reveal where the 4 originally appeared? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Learned representations (ai-5/19). Goal: Trace an input through a hidden representation and a decision rule. Short practice: Why can no deterministic head using only the sum classify both correctly? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
